Effective two-component and two-mediator dark matter

Diego Restrepo

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Phenomenology Group
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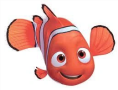
In collaboration with
Kimy Agudelo, Andrés Rivera and David Suárez (UdeA)



Effective neutrino masses

Majorana

Baryon and Lepton Nonconserving Processes #1

Steven Weinberg (Harvard U.) (1979)

Published in: *Phys.Rev.Lett.* 43 (1979) 1566-1570

Dirac

Systematic Study of One-Loop Dirac Neutrino Masses and Viable Dark Matter Candidates #1

Chang-Yuan Yao (Hefei, CUST), Gui-Jun Ding (Hefei, CUST) (Jul 31, 2017)

Published in: *Phys.Rev.D* 96 (2017) 9, 095004, *Phys.Rev.D* 98 (2018) 3, 039901 (erratum) • e-Print: 1707.09786 [hep-ph]

Name	$SU(2)_L$	$U(1)_Y$
L	$\mathbf{2}$	$-1/2$
H	$\mathbf{2}$	$1/2$

$$\mathcal{L}_M = L \cdot H L \cdot H + \text{h.c}$$

$$\mathcal{L}_D = (\nu_R)^\dagger L \cdot H + \text{h.c}$$

Dark matter (radiative) realization $\rightarrow Z_2$

$$\cancel{M_{N \nu_R \nu_R}} \rightarrow U(1)$$

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Scotogenic one-loop realizations

Majorana

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Systematic study of the d=5 Weinberg operator at one-loop order #1

Florian Bonnet (Wurzburg U.), Martin Hirsch (Valencia U., IFIC), Toshihiko Ota (Munich, Max Planck Inst.), Walter Winter (Wurzburg U.) (Apr, 2012)

Published in: *JHEP* 07 (2012) 153 • e-Print: 1204.5862 [hep-ph]

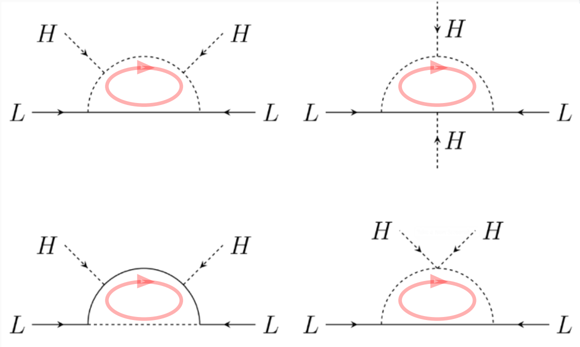
pdf DOI cite claim reference search 197 citations

Models with radiative neutrino masses and viable dark matter candidates #1

Diego Restrepo (Antioquia U.), Oscar Zapata (Antioquia U.), Carlos E. Yaguna (Munster U., ITP) (Aug 16, 2013)

Published in: *JHEP* 11 (2013) 011 • e-Print: 1308.3655 [hep-ph]

pdf DOI cite claim reference search 123 citations

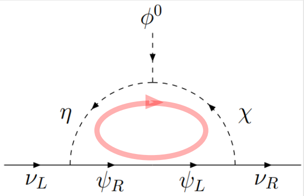


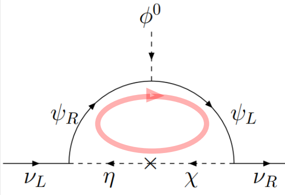
Pathways to Naturally Small Dirac Neutrino Masses #1

Ernest Ma (UC, Riverside), Oleg Popov (UC, Riverside) (Sep 8, 2016)

Published in: *Phys.Lett.B* 764 (2017) 142-144 • e-Print: 1609.02538 [hep-ph]

pdf DOI cite claim reference search 125 citations





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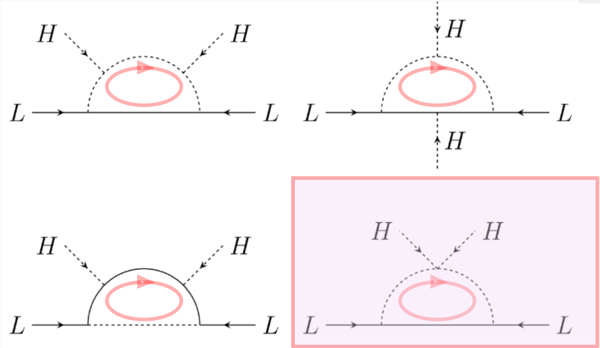

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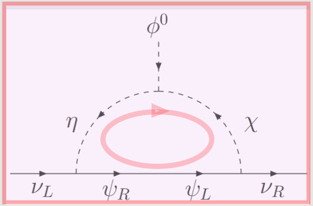
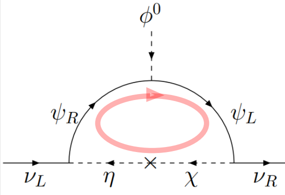




Majorana

Dirac

Radiative seesaw mechanism at weak scale #1

Zhi-jian Tao (Beijing, Inst. High Energy Phys.) (Mar, 1996)

Published in: *Phys.Rev.D* 54 (1996) 5693-5697 • e-Print: hep-ph/9603309 [hep-ph]

pdf DOI cite claim reference search 34 citations

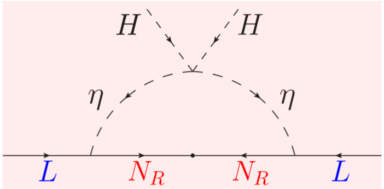
Verifiable radiative seesaw mechanism of neutrino mass and dark matter #2

Ernest Ma (UC, Riverside) (Jan, 2006)

Published in: *Phys.Rev.D* 73 (2006) 077301 • e-Print: hep-ph/0601225 [hep-ph]

pdf DOI cite claim reference search 1,539 citations

Name	$SU(2)_L$	$U(1)_Y$	Z_2
N_R	**1**	0	$-$
η	**2**	$1/2$	$-$



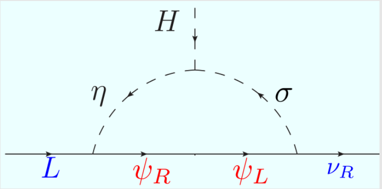
Radiative Neutrino Mass, Dark Matter and Leptogenesis #1

Pei-Hong Gu (ICTP, Trieste), Utpal Sarkar (Ahmedabad, Phys. Res. Lab) (Dec, 2007)

Published in: *Phys.Rev.D* 77 (2008) 105031 • e-Print: 0712.2933 [hep-ph]

pdf DOI cite claim reference search 124 citations

Name	$SU(2)_L$	$U(1)_Y$	Z_2	$U(1)_L$
ν_R	**1**	0	$+$	-1
ψ_R	**1**	0	$-$	-1
ψ_L	**1**	0	$-$	-1
η	**2**	$1/2$	$-$	0
σ	**1**	0	$-$	0



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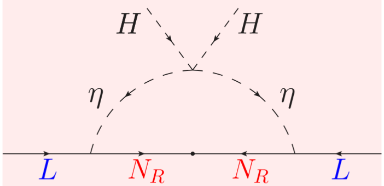

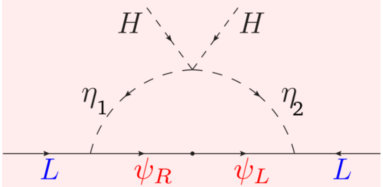

New Scotogenic Model of Neutrino Mass with $U(1)_D$ Gauge Interaction

#1

Ernest Ma (UC, Riverside), Ivica Picek (Zagreb U., Phys. Dept.), Branimir Radovčić (Zagreb U., Phys. Dept.) (Aug 24, 2013)

Published in: *Phys.Lett.B* 726 (2013) 744-746 • e-Print: 1308.5313 [hep-ph]

pdf DOI cite claim reference search 86 citations

Dirac

Radiative Neutrino Mass, Dark Matter and Leptogenesis

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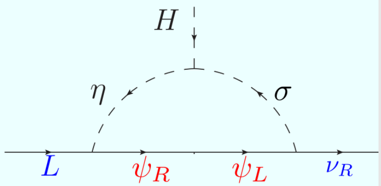
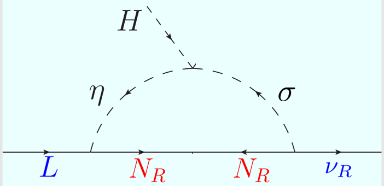
Pei-Hong Gu (ICTP, Trieste), Utpal Sarkar (Ahmedabad, Phys. Res. Lab) (Dec, 2007)

Published in: *Phys.Rev.D* 77 (2008) 105031 • e-Print: 0712.2933 [hep-ph]

pdf DOI cite claim reference search 124 citations





Dirac neutrino mass generation from a Majorana messenger

#1

Julian Calle (Antioquia U.), Diego Restrepo (Antioquia U. and IIP, Brazil), Óscar Zapata (Antioquia U.) (Sep 20, 2019)

Published in: *Phys.Rev.D* 101 (2020) 3, 035004 • e-Print: 1909.09574 [hep-ph]

pdf links DOI cite claim reference search 15 citations

From $Z_2 \times U(1)$ to local $U(1)_{B-L}$: Dirac

Dark matter stability and Dirac neutrinos using only Standard Model symmetries #1

Cesar Bonilla (Munich, Tech. U.), Salvador Centelles-Chuliá (Valencia U., IFIC), Ricardo Cepedello (Valencia U., IFIC), Eduardo Peinado (Mexico U.), Rahul Srivastava (Valencia U., IFIC) (Dec 4, 2018)

Published in: *Phys.Rev.D* 101 (2020) 3, 033011 • e-Print: 1812.01599 [hep-ph]

Minimal radiative Dirac neutrino mass models #1

Julian Calle (Antioquia U.), Diego Restrepo (Antioquia U.), Carlos E. Yaguna (UPTC, Tunja), Óscar Zapata (Antioquia U.) (Dec 13, 2018)

Published in: *Phys.Rev.D* 99 (2019) 7, 075008 • e-Print: 1812.05523 [hep-ph]

$$\mathcal{L}_5 = \frac{h}{\Lambda}(\nu_R)^\dagger L \cdot H S^* + \text{H.c.},$$

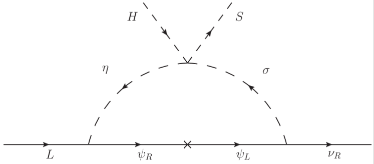


FIG. 6. T3-1-A-I ($\alpha = 0$): $B - L$ flux in the Dirac radiative seesaw.

T3-1-A-I	$(\nu_{Ri})^\dagger$	$(\nu_{Rj})^\dagger$	$(\nu_{Rk})^\dagger$	$(\psi_R)^\dagger$	ψ_L	η_a	σ_a	—	S
L	$+4$	$+4$	-5	r	$-r$	$1-r$	$4-r$	—	3

Anomaly cancellation of a U(1) gauge symmetry with chiral fermions

In a fundamental theory the elementary fermions are expected to be chiral, i.e, the their charges should not allow a mass term larger than the scale of spontaneous symmetry breaking. For any set of chargues associted to a U(1) gauge symmetry

$$\boldsymbol{Z} = [Z_1, Z_2, \cdots, Z_N] \,,$$

The triangle anomaly with three U(1) gauge bosons on the external lines, i.e., the $[U(1)]^3$ anomaly, and with one U(1) gauge boson and two gravitons on the external lines should also be cancelled

$$\sum_{\alpha=1}^{N} Z_\alpha = 0\,, \qquad \sum_{\alpha=1}^{N} Z_\alpha^3 = 0\,, \tag{1}$$

$U(1)_Y$: no vector-like, i.e., pairs of fields with 'equal-but-opposite' charges

The hypercharge for one generation in the standard model can be seen as gauge Abelian symmetry $U(1)_Y$ with 15 left-handed Weyl massless fermions with integer charges

$$Y = [1, 1, 1, 1, 1, 1, -4, -4, -4, 2, 2, 2, -3, -3, 6]$$

which satisfy

$$\sum_i Y_i = 0\,, \qquad \sum_i Y_i^3 = 0\,.$$

If we introduce an scalar, H, of charge 3 we can form Yukawa couplings through the Dirac pairs

$$u_\alpha = (1, -4), (1, -4), (1, -4), \qquad d_\alpha = (1, 2), (1, 2), (1, 2), \qquad e = (-3, 6)$$

and the *chiral fermions* acquire Dirac masses after the EWSB. They are degenerate because the additional color symmetry. One remains massless, $\nu_L = (-3)$. The renmant Z_3 symmetry guarantees the stability of the lightest quark

Secluded gauge $U(1)_D$ without vector-like fermions:

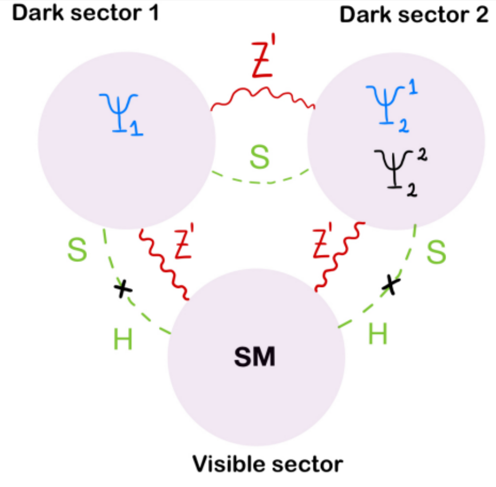
$$\boldsymbol{S} = [\chi_1, \chi_2, \cdots, \psi_1, \psi_2, \cdots, \psi_{N'}]$$

- *Dark Higgs mechanism*: Singlet dark Higgs ϕ acquires a vev and give mass to the dark photon

$$\mathcal{L} = i\psi_a^\dagger \overline{\sigma^\mu}\left(\partial_\mu - ig_D Z_\mu^D\right)\psi_a - \frac{1}{4}V_{\mu\nu}V^{\mu\nu} + \sum_{a<b} h_{ab}\psi_a\psi_b\phi^{(*)} + \text{h.c} - V(\phi)\,. \qquad (2)$$

- S_α are the charges of SM-singlet right-handed chiral fermions with $N \geq 5$
  - χ_i *massless fermions* with $i = 1, \cdots, N'$ with $N' \leq N$
  - ψ_a *multi-component dark matter*: massive after the spontaneous symmetry breaking of $U(1)_D$ with $a = N' + 1, \cdots, N$
- *Larger parameter space*: Dark photon, Z_D, exclusions instead of Z'
- Z_D and ϕ masses are related by (arXiv:1610.03063)

$$h/g_D = \sqrt{2}m_\psi/m_{Z_D}\,.$$

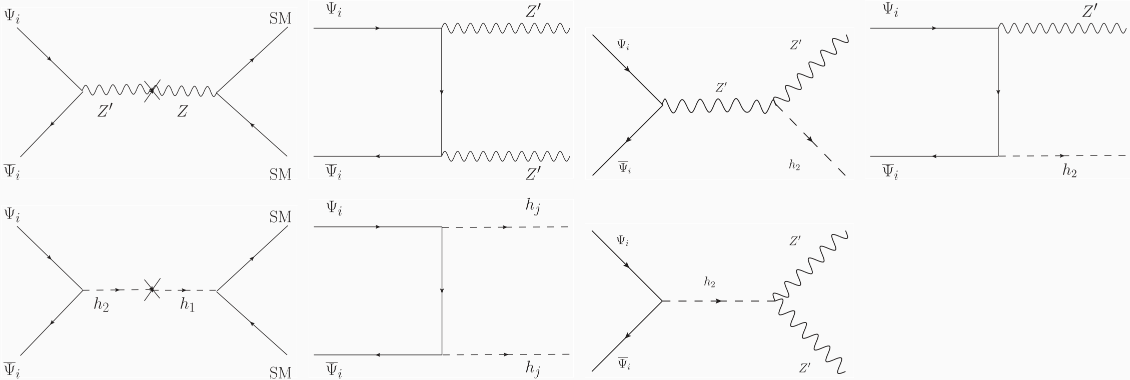

Dark sector 1
Dark sector 2
Z'
Ψ_1
Ψ_2^1
S
Ψ_2^2
S
Z'
Z'
S
H
H
SM
Visible sector

Two-mediator: Dark photon and dark Higgs

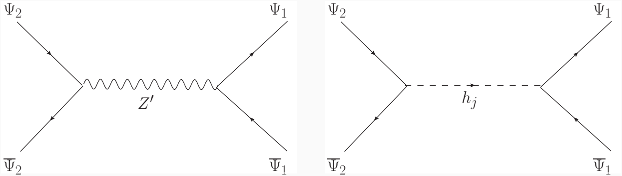
$$\begin{pmatrix} h_1 \\ h_2 \end{pmatrix} = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} \mathrm{Re}(H^0) \\ \mathrm{Re}(S) \end{pmatrix}$$

$$\mathcal{L}_{B'} \supset -\frac{\epsilon}{2} B'_{\mu\nu} B^{\mu\nu}$$

$$\begin{pmatrix} A^\mu \\ Z^\mu \\ Z'^{\,\mu} \end{pmatrix} = \frac{1}{4} v^2 \begin{pmatrix} g_1^2 & -g_1 g_2 & -g_1^2 \epsilon \\ -g_1 g_2 & g_2^2 & g_1 g_2 \epsilon \\ -g_1^2 \epsilon & g_1 g_2 \epsilon & 324 g_D^2 \frac{v_S^2}{v^2} + g_1^2 \epsilon^2 \end{pmatrix} \begin{pmatrix} B^\mu \\ W_3^\mu \\ B'^{\,\mu} \end{pmatrix}$$



DM conversion channels



Decrease the number of charges to be assigned to dark matter particles, ψ_i below

$$[\chi_1, \chi_2, \cdots, \psi_1, \psi_2, \cdots, \psi_{N'}]$$

Secluded case:

$$[\nu, \nu, (\nu), \psi_1, \psi_2, \cdots, \psi_{N'}]$$

$$\chi_1 \to \nu_{R1}, \cdots, \chi_{N_\nu} \to \nu_{R\,N_\nu}, \qquad 2 \leq N_\nu \leq 3\,,$$

$$\mathcal{L}_{\text{eff}} = h_\nu^{ij}\,(\nu_{Ri})^\dagger\,\epsilon_{ab}\,L_j^a\,H^b\left(\frac{\phi^*}{\Lambda}\right)^\delta + \text{H.c.}, \qquad \text{with } i,j = 1,2,3\,,$$

ϕ is the complex singlet scalar responsible for the SSB of the anomaly-free gauge symmetry and $\boxed{\text{give mass to } \textit{all } \psi_a}$

$$\phi = -\frac{\nu}{\delta}\,,$$

Effective Dirac neutrino mass operator

Decrease the number of charges to be assigned to dark matter particles, ψ_i below

$$[\chi_1, \chi_2, \cdots, \psi_1, \psi_2, \cdots, \psi_{N'}]$$

Secluded case:

$$[5, 5, -3, -2, 1, -6]$$

$$\chi_1 \to \nu_{R1}, \; \chi_2 \to \nu_{R2}, \qquad N_\nu = 2\,,$$

$$\mathcal{L}_{\text{eff}} = h_\nu^{aj} \, (\nu_{Ra})^\dagger \, \epsilon_{bc} \, L_j^b \, H^c \left(\frac{\phi^*}{\Lambda}\right) + \text{H.c.}, \qquad \text{with } j = 1, 2, 3\,,$$

Decrease the number of charges to be assigned to dark matter particles, ψ_i below

$$[\chi_1, \chi_2, \cdots, \psi_1, \psi_2, \cdots, \psi_{N'}]$$

Secluded case:

$$[5, 5, -3, -2, 1, -6]$$

$$\chi_1 \to \nu_{R1},\ \chi_2 \to \nu_{R2}, \qquad N_\nu = 2\,,$$

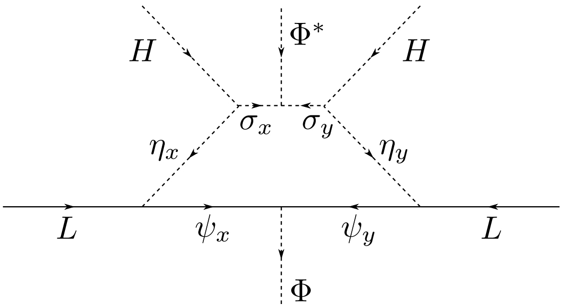
$$\mathcal{L}_{\text{eff}} = h_\nu^{aj}\,(\nu_{Ra})^\dagger\,\epsilon_{bc}\,L_j^b\,H^c\left(\frac{\phi^*}{\Lambda}\right) + \text{H.c.}, \qquad \text{with } j = 1, 2, 3\,,$$

$$\boldsymbol{z} = [5, 5, -3, -2, 1, -6] \to \phi = -5 \to [(5, 5), (-3, -2), (1, -6)]$$

$$\mathcal{L} \subset h_{(-3,-2)}\psi_{-3}\psi_{-2}\phi^* + h_{(1,-5)}\psi_1\psi_{-6}\phi^* + \text{h.c.}$$

Majorana

$[\psi_1, \psi_2, \cdots, \psi_N]$

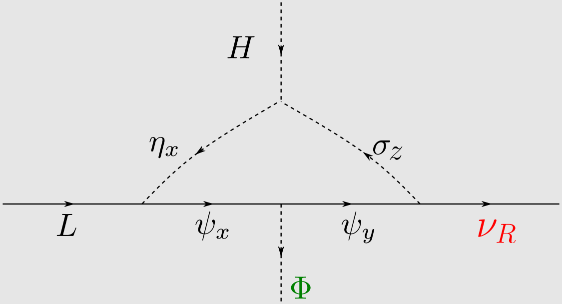


$$\frac{y}{\Lambda} LLHH \to \frac{y}{\Lambda} LLHH \frac{\phi}{\Lambda} \frac{\phi^*}{\Lambda}$$

ϕ give mass to *all* ψ_a

Dirac

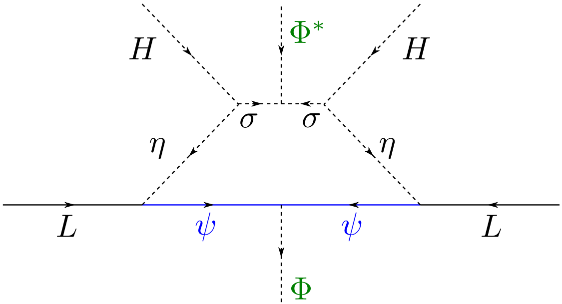
$[\nu, \nu, (\nu), \psi_1, \psi_2, \cdots, \psi_{N'}]$



$$y (\nu_R)^\dagger LH \to y (\nu_R)^\dagger LH \frac{\phi^*}{\Lambda}$$

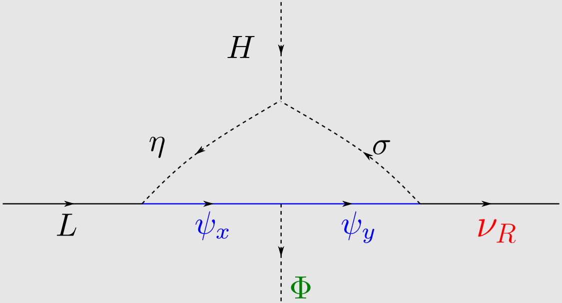
ϕ give mass to *all* ψ_a

$$[\psi_1, \psi_2, \cdots, \psi_N]$$



$$\phi = 2 \to [\underbrace{1, 1}_{\psi_a}, (2, -4), (4, -6), (3, -5)]$$

$$[\nu, \nu, (\nu), \psi_1, \psi_2, \cdots, \psi_{N'}]$$



$$\phi = 9 \to [[9, 9, 9, \underbrace{(1, -10), (1, -10)}_{\psi_a}, (-4, -5)],$$

Anomaly-free chiral $U(1)_D$ and its scotogenic implication #1

Chi-Fong Wong (Macau U. Sci. Tech., SSI) (Aug 19, 2020)

Published in: *Phys.Dark Univ.* 32 (2021) 100818 • e-Print: 2008.08573 [hep-ph]

Anomaly-free Abelian gauge symmetries with Dirac scotogenic models #8

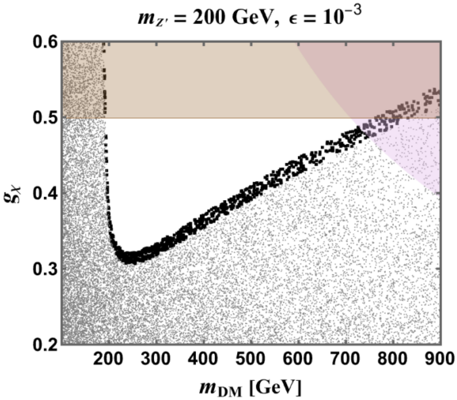
Nicolás Bernal (Antonio Narino U.), Julián Calle (Antioquia U.), Diego Restrepo (Antioquia U.) (Feb 11, 2021)

Published in: *Phys.Rev.D* 103 (2021) 9, 095032 • e-Print: 2102.06211 [hep-ph]

Majorana — Minimal inert scalar sector. — Dirac

$\phi = 2 \rightarrow [1, 1, (2, -4), (4, -6), (3, -5)]$

Secluded DM with Majorana fermion candidate



Chiral dark matter and radiative neutrino masses from gauged U(1) symmetry #1

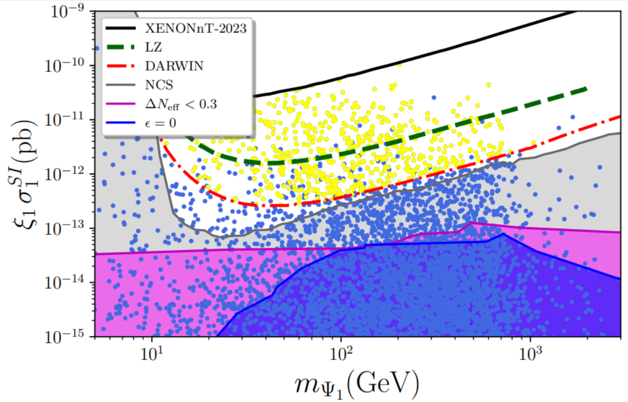
K.S. Babu (Oklahoma State U.), Shreyashi Chakdar (Holy Cross Coll. and Munster U., ITP), Vishnu P. K (Sep 13, 2024)

e-Print: 2409.09008 [hep-ph]

pdf cite claim reference search 0 citations

$\phi = 9 \rightarrow [[9, 9, 9, (1, -10), (1, -10), (-4, -5)],$

New decay channel: $Z_D \rightarrow \bar{\nu}\nu$



In progress...

Many other venues

$$m_\nu \propto \frac{\langle H^0 \rangle^2}{\Lambda} \times \left(\frac{1}{16\pi^2} \right)^n \times \epsilon \times \left(\frac{\langle H^0 \rangle}{\Lambda} \right)^{d-5},$$

Other representa-tions

Other Topologies

Two or three loops

Extra coupling suppression

Effective operators of higher dimension